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Reversible valve for HVAC system

Abstract

A heating, ventilation, and/or air conditioning (HVAC) system includes a reversible valve having an outlet configured to direct refrigerant to a reheat heat exchanger of the HVAC system. The reversible valve is further configured to be in a first configuration to direct the refrigerant through a refrigerant circuit in a first flow direction in a heating mode of the HVAC system and to be in a second configuration to direct the refrigerant through the refrigerant circuit in a second flow direction in a cooling mode of the HVAC system.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 16/591,412, entitled “REVERSIBLE VALVE FOR HVAC SYSTEM,” filed Oct. 2, 2019, which claims priority from and the benefit of U.S. Provisional Application No. 62/852,883, entitled “REVERSIBLE VALVE FOR HVAC SYSTEM,” filed May 24, 2019, each of which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

(1) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, and are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

(2) Heating, ventilation, and air conditioning (HVAC) systems are utilized in residential, commercial, and industrial environments to control environmental properties, such as temperature and humidity, for occupants of the respective environments. An HVAC system may control the environmental properties through control of an air flow delivered to the environment. For example, the HVAC system may place the air flow in a heat exchange relationship with a refrigerant of a vapor compression circuit to condition the air flow. In some embodiments, the vapor compression circuit may include a reheat heat exchanger that is used to control a moisture content of the air flow to achieve a target humidity of the air flow. The HVAC system may also include a supplemental heat source may be used to heat the air flow. However, including the supplemental heat source may increase a cost associated with manufacturing and/or operating the HVAC system. In alternative embodiments, the vapor compression circuit may be configured to operate in a cooling mode to cool the air flow and a heating mode to heat the air flow. However, such embodiments may not have a reheat heat exchanger and, as a result, may have a limited capability of controlling the moisture content of the air flow.

SUMMARY

(3) A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(4) In one embodiment, heating, ventilation, and/or air conditioning (HVAC) system configured to condition an air flow includes a reversible valve that includes a first port, wherein the reversible valve is configured to be in a first configuration to direct refrigerant through a refrigerant circuit of the HVAC system in a first flow direction in a heating mode of the HVAC system and to be in a second configuration to direct the refrigerant through the refrigerant circuit in a second flow direction in a cooling mode of the HVAC system. In addition, the HVAC system includes a reheat heat exchanger configured to transfer heat from the refrigerant to the air flow in the heating mode and in the cooling mode, wherein the reversible valve is configured to discharge the refrigerant to the reheat heat exchanger via the first port in the heating mode and in the cooling mode.

(5) In another embodiment, a heating, ventilation, and/or air conditioning (HVAC) system includes a compressor including a compressor discharge and a reheat heat exchanger configured to transfer heat from refrigerant to an air flow directed through the HVAC system in a heating mode of the HVAC system and in a cooling mode of the HVAC system, wherein the reheat heat exchanger is configured to receive the air flow from an additional heat exchanger of the HVAC system. The HVAC system also includes a reversing valve comprising an inlet configured to receive the

refrigerant from the compressor discharge, a first port configured to discharge the refrigerant to the reheat heat exchanger in the heating mode and in the cooling mode, and a second port configured to discharge the refrigerant to the additional heat exchanger in the heating mode and to receive the refrigerant from the additional heat exchanger in the cooling mode.

(6) In another embodiment, a heat pump configured to operate in a heating mode and in a cooling mode includes a reheat heat exchanger configured to transfer heat from refrigerant of the heat pump to an air flow supplied to a space within a building in the heating mode and in the cooling mode. In addition, the heat pump includes a first heat exchanger configured to transfer heat between refrigerant of the heat pump and the air flow supplied to the space within the building, a reversible valve having an inlet configured to receive pressurized refrigerant from a compressor of the heat pump, wherein the reversible valve is configured to be in a first configuration in the heating mode of the heat pump and to be in a second configuration in the cooling mode of the heat pump, and a controller configured to cause actuation of the reversible valve between the first configuration and the second configuration, and the reversible valve is configured to discharge the pressurized refrigerant to the reheat heat exchanger via a first port in the first configuration and the second configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:

(2) FIG. 1 is a perspective view of an embodiment of a heating, ventilation, and/or air conditioning (HVAC) system for environmental management that may employ one or more HVAC units, in accordance with an aspect of the present disclosure;

(3) FIG. 2 is a perspective view of an embodiment of a packaged HVAC unit that may be used in the HVAC system of FIG. 1, in accordance with an aspect of the present disclosure;

(4) FIG. 3 is a cutaway perspective view of an embodiment of a residential, split HVAC system, in accordance with an aspect of the present disclosure;

(5) FIG. 4 is a schematic of an embodiment of a vapor compression system that can be used in any of the systems of FIGS. 1-3, in accordance with an aspect of the present disclosure, in accordance with an aspect of the present disclosure;

(6) FIG. 5 is a schematic of an embodiment of an HVAC system having a reversible valve in a heating configuration to operate the HVAC system in a heating mode with reheat functionality, in accordance with an aspect of the present disclosure;

(7) FIG. 6 is a schematic of an embodiment of an HVAC system having a reversible valve in a cooling configuration to operate the HVAC system in a cooling mode with reheat functionality, in accordance with an aspect of the present disclosure;

(8) FIG. 7 is a schematic of an embodiment of an HVAC system having a reversible valve and an additional heat exchanger shared by an additional HVAC system, in accordance with an aspect of the present disclosure;

(9) FIG. 8 is a schematic of an embodiment of an HVAC system having a reversible valve configured to operate the HVAC system in a heating mode with reheat functionality or a cooling mode with reheat functionality, in accordance with an aspect of the present disclosure;

(10) FIG. 9 is a schematic of an embodiment of a reversible valve fluidly coupled to a pilot valve positioning the reversible valve in a cooling configuration, in accordance with an aspect of the present disclosure; and

(11) FIG. 10 is a schematic of an embodiment of a reversible valve fluidly coupled to a pilot valve positioning the reversible valve in a heating configuration, in accordance with an aspect of the

present disclosure.

DETAILED DESCRIPTION

(12) One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be noted that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(13) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(14) The present disclosure is directed to a heating, ventilation, and/or air conditioning (HVAC) system configured to condition an air flow and to supply the conditioned air flow to a structure or building. To this end, the HVAC system may receive an air flow, which may be referred to as an entering air flow, and may include a return air flow from the building and/or an outdoor air flow from an ambient environment. The HVAC system may include a first heat exchanger configured to place the air flow in a heat exchange relationship with refrigerant flowing through the first heat exchanger. Thermal energy, such as heat, may transfer between the air flow and the refrigerant to condition the air flow. The HVAC system may then deliver the conditioned air flow as supply air flow to the building to condition a space within the building. The HVAC system may also include a second heat exchanger configured to change a temperature of the refrigerant before and after the refrigerant flows through the first heat exchanger. For example, the second heat exchanger may place the refrigerant in a heat exchange relationship with an outdoor ambient air flow in order to change the temperature of the refrigerant to enable the refrigerant to condition the air flow.

(15) In some embodiments, the HVAC system may include a reheat heat exchanger that may be used to control the temperature and the humidity of the supply air flow. For instance, the HVAC system may use the first heat exchanger to cool the entering air flow and remove a certain amount of moisture from the entering air flow to achieve a target humidity of the generated supply air flow. The HVAC system may then use the reheat heat exchanger to heat the entering air flow to a target temperature, and then deliver the entering air flow to the structure as the supply air flow. In conventional systems, the HVAC system may include a refrigerant circuit, also referred to as a vapor compression circuit, which is configured to operate in a cooling mode or a reheat mode but not a heating mode. Thus, a supplemental heat source, such as a furnace system, may be implemented to heat the entering air flow and enable the HVAC system to heat a space within the building. However, the supplemental heat source may increase a cost associated with manufacturing and/or operating the HVAC system.

(16) In alternative systems, the HVAC system may be a heat pump having a refrigerant circuit configured to operate in a cooling mode and in a heating mode. However, conventional heat pumps may not have a reheat heat exchanger. For this reason, the heat pump system may not sufficiently control the humidity of supply air delivered to a building, thereby limiting the functionality of the heat pump.

(17) Thus, it is presently recognized that a refrigerant circuit configured to operate in a cooling mode and in a heating mode and that also includes a reheat heat exchanger may improve the

functionality and performance of the HVAC system, while also reducing costs associated with HVAC system manufacture, operation, and maintenance. Accordingly, embodiments of the present disclosure are directed to an HVAC system having a heat pump with a reheat heat exchanger and a reversible or a reversing valve configured to enable cooling, heating, and reheat operations.

(18) The reversible valve may be configured to alternate between a first configuration and a second configuration. In the first configuration, the reversible valve may cause the refrigerant to flow through a refrigerant circuit of the heat pump along a first flow path or direction to enable the heat pump to operate in a heating mode. In the second configuration, the reversible valve may cause the refrigerant to flow through the heat pump along a second flow path or direction of the refrigerant circuit to enable the heat pump to operate in a cooling mode. Further, in both the first configuration and the second configuration, the reversible valve may enable the refrigerant to flow through the reheat heat exchanger disposed along the refrigerant circuit, which places the refrigerant in another heat exchange relationship with the entering air flow.

(19) The functionality of the reheat exchanger may depend on whether the heat pump is in the cooling mode or the heating mode. For example, in the heating mode, the reheat heat exchanger may be used to provide supplemental heating of the supply air flow. Thus, the reheat heat exchanger enables the heat pump to heat the space within the building more efficiently. Additionally, in the cooling mode, the reheat heat exchanger may be used to facilitate dehumidification of the entering air flow in order to achieve a target humidity and/or temperature of the supply air flow. As such, a supplemental heat source may not be utilized to provide heating capabilities, thereby reducing a cost to manufacture and/or to operate the heat pump. The reversible valve may also be used to adjust the amount of refrigerant directed to the reheat heat exchanger during the heating and/or cooling mode, thereby controlling an amount of supplemental heat or reheat provided by the reheat heat exchanger to the air flow. In this manner, the reversible valve enables greater control of the heat pump to condition the entering air flow, so as to achieve the target temperature and/or humidity of the supply air flow more accurately and/or efficiently.

(20) Turning now to the drawings, FIG. 1 illustrates an embodiment of a heating, ventilation, and/or air conditioning (HVAC) system for environmental management that may employ one or more HVAC units. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an “HVAC system” as used herein is defined as conventionally understood and as further described herein. Components or parts of an “HVAC system” may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An “HVAC system” is a system configured to provide such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

(21) In the illustrated embodiment, a building **10** is air conditioned by a system that includes an HVAC unit **12**. The building **10** may be a commercial structure or a residential structure. As shown, the HVAC unit **12** is disposed on the roof of the building **10**; however, the HVAC unit **12** may be located in other equipment rooms or areas adjacent the building **10**. The HVAC unit **12** may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit **12** may be part of a split HVAC system, such as the system shown in FIG. 3, which includes an outdoor HVAC unit **58** and an indoor HVAC unit **56**.

(22) The HVAC unit **12** is an air cooled device that implements a refrigeration cycle to provide

conditioned air to the building **10**. Specifically, the HVAC unit **12** may include one or more heat exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building. In the illustrated embodiment, the HVAC unit **12** is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building **10**. After the HVAC unit **12** conditions the air, the air is supplied to the building **10** via ductwork **14** extending throughout the building **10** from the HVAC unit **12**. For example, the ductwork **14** may extend to various individual floors or other sections of the building **10**. In certain embodiments, the HVAC unit **12** may be a heat pump that provides both heating and cooling to the building with one refrigeration circuit configured to operate in different modes. In other embodiments, the HVAC unit **12** may include one or more refrigeration circuits for cooling an air stream and a furnace for heating the air stream.

(23) A control device **16**, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air. The control device **16** also may be used to control the flow of air through the ductwork **14**. For example, the control device **16** may be used to regulate operation of one or more components of the HVAC unit **12** or other components, such as dampers and fans, within the building **10** that may control flow of air through and/or from the ductwork **14**. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of the supply air, return air, and so forth. Moreover, the control device **16** may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building **10**.

(24) FIG. 2 is a perspective view of an embodiment of the HVAC unit **12**. In the illustrated embodiment, the HVAC unit **12** is a single package unit that may include one or more independent refrigeration circuits and components that are tested, charged, wired, piped, and ready for installation. The HVAC unit **12** may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with gas heat, or cooling with a heat pump. As described above, the HVAC unit **12** may directly cool and/or heat an air stream provided to the building **10** to condition a space in the building **10**.

(25) As shown in the illustrated embodiment of FIG. 2, a cabinet **24** encloses the HVAC unit **12** and provides structural support and protection to the internal components from environmental and other contaminants. In some embodiments, the cabinet **24** may be constructed of galvanized steel and insulated with aluminum foil faced insulation. Rails **26** may be joined to the bottom perimeter of the cabinet **24** and provide a foundation for the HVAC unit **12**. In certain embodiments, the rails **26** may provide access for a forklift and/or overhead rigging to facilitate installation and/or removal of the HVAC unit **12**. In some embodiments, the rails **26** may fit into “curbs” on the roof to enable the HVAC unit **12** to provide air to the ductwork **14** from the bottom of the HVAC unit **12** while blocking elements such as rain from leaking into the building **10**.

(26) The HVAC unit **12** includes heat exchangers **28** and **30** in fluid communication with one or more refrigeration circuits. Tubes within the heat exchangers **28** and **30** may circulate refrigerant, such as R-410A, through the heat exchangers **28** and **30**. The tubes may be of various types, such as multichannel tubes, conventional copper or aluminum tubing, and so forth. Together, the heat exchangers **28** and **30** may implement a thermal cycle in which the refrigerant undergoes phase changes and/or temperature changes as it flows through the heat exchangers **28** and **30** to produce heated and/or cooled air. For example, the heat exchanger **28** may function as a condenser where heat is released from the refrigerant to ambient air, and the heat exchanger **30** may function as an evaporator where the refrigerant absorbs heat to cool an air stream. In other embodiments, the HVAC unit **12** may operate in a heat pump mode where the roles of the heat exchangers **28** and **30** may be reversed. That is, the heat exchanger **28** may function as an evaporator and the heat exchanger **30** may function as a condenser. In further embodiments, the HVAC unit **12** may include a furnace for heating the air stream that is supplied to the building **10**. While the illustrated

embodiment of FIG. 2 shows the HVAC unit **12** having two of the heat exchangers **28** and **30**, in other embodiments, the HVAC unit **12** may include one heat exchanger or more than two heat exchangers.

(27) The heat exchanger **30** is located within a compartment **31** that separates the heat exchanger **30** from the heat exchanger **28**. Fans **32** draw air from the environment through the heat exchanger **28**. Air may be heated and/or cooled as the air flows through the heat exchanger **28** before being released back to the environment surrounding the HVAC unit **12**. A blower assembly **34**, powered by a motor **36**, draws air through the heat exchanger **30** to heat or cool the air. The heated or cooled air may be directed to the building **10** by the ductwork **14**, which may be connected to the HVAC unit **12**. Before flowing through the heat exchanger **30**, the conditioned air flows through one or more filters **38** that may remove particulates and contaminants from the air. In certain embodiments, the filters **38** may be disposed on the air intake side of the heat exchanger **30** to prevent contaminants from contacting the heat exchanger **30**.

(28) The HVAC unit **12** also may include other equipment for implementing the thermal cycle. Compressors **42** increase the pressure and temperature of the refrigerant before the refrigerant enters the heat exchanger **28**. The compressors **42** may be any suitable type of compressors, such as scroll compressors, rotary compressors, screw compressors, or reciprocating compressors. In some embodiments, the compressors **42** may include a pair of hermetic direct drive compressors arranged in a dual stage configuration **44**. However, in other embodiments, any number of the compressors **42** may be provided to achieve various stages of heating and/or cooling. Additional equipment and devices may be included in the HVAC unit **12**, such as a solid-core filter drier, a drain pan, a disconnect switch, an economizer, pressure switches, phase monitors, and humidity sensors, among other things.

(29) The HVAC unit **12** may receive power through a terminal block **46**. For example, a high voltage power source may be connected to the terminal block **46** to power the equipment. The operation of the HVAC unit **12** may be governed or regulated by a control board **48**. The control board **48** may include control circuitry connected to a thermostat, sensors, and alarms. One or more of these components may be referred to herein separately or collectively as the control device **16**. The control circuitry may be configured to control operation of the equipment, provide alarms, and monitor safety switches. Wiring **49** may connect the control board **48** and the terminal block **46** to the equipment of the HVAC unit **12**.

(30) FIG. 3 illustrates a residential heating and cooling system **50**, also in accordance with present techniques. The residential heating and cooling system **50** may provide heated and cooled air to a residential structure, as well as provide outside air for ventilation and provide improved indoor air quality (IAQ) through devices such as ultraviolet lights and air filters. In the illustrated embodiment, the residential heating and cooling system **50** is a split HVAC system. In general, a residence **52** conditioned by a split HVAC system may include refrigerant conduits **54** that operatively couple the indoor unit **56** to the outdoor unit **58**. The indoor unit **56** may be positioned in a utility room, an attic, a basement, and so forth. The outdoor unit **58** is typically situated adjacent to a side of residence **52** and is covered by a shroud to protect the system components and to prevent leaves and other debris or contaminants from entering the unit. The refrigerant conduits **54** transfer refrigerant between the indoor unit **56** and the outdoor unit **58**, typically transferring primarily liquid refrigerant in one direction and primarily vaporized refrigerant in an opposite direction.

(31) When the system shown in FIG. 3 is operating as an air conditioner, a heat exchanger **60** in the outdoor unit **58** serves as a condenser for re-condensing vaporized refrigerant flowing from the indoor unit **56** to the outdoor unit **58** via one of the refrigerant conduits **54**. In these applications, a heat exchanger **62** of the indoor unit functions as an evaporator. Specifically, the heat exchanger **62** receives liquid refrigerant, which may be expanded by an expansion device, and evaporates the refrigerant before returning it to the outdoor unit **58**.

(32) The outdoor unit **58** draws environmental air through the heat exchanger **60** using a fan **64** and expels the air above the outdoor unit **58**. When operating as an air conditioner, the air is heated by the heat exchanger **60** within the outdoor unit **58** and exits the unit at a temperature higher than it entered. The indoor unit **56** includes a blower or fan **66** that directs air through or across the indoor heat exchanger **62**, where the air is cooled when the system is operating in air conditioning mode. Thereafter, the air is passed through ductwork **68** that directs the air to the residence **52**. The overall system operates to maintain a desired temperature as set by a system controller. When the temperature sensed inside the residence **52** is higher than the set point on the thermostat, or the set point plus a small amount, the residential heating and cooling system **50** may become operative to refrigerate additional air for circulation through the residence **52**. When the temperature reaches the set point, or the set point minus a small amount, the residential heating and cooling system **50** may stop the refrigeration cycle temporarily.

(33) The residential heating and cooling system **50** may also operate as a heat pump. When operating as a heat pump, the roles of heat exchangers **60** and **62** are reversed. That is, the heat exchanger **60** of the outdoor unit **58** will serve as an evaporator to evaporate refrigerant and thereby cool air entering the outdoor unit **58** as the air passes over the outdoor heat exchanger **60**. The indoor heat exchanger **62** will receive a stream of air blown over it and will heat the air by condensing the refrigerant.

(34) In some embodiments, the indoor unit **56** may include a furnace system **70**. For example, the indoor unit **56** may include the furnace system **70** when the residential heating and cooling system **50** is not configured to operate as a heat pump. The furnace system **70** may include a burner assembly and heat exchanger, among other components, inside the indoor unit **56**. Fuel is provided to the burner assembly of the furnace **70** where it is mixed with air and combusted to form combustion products. The combustion products may pass through tubes or piping in a heat exchanger, separate from heat exchanger **62**, such that air directed by the blower **66** passes over the tubes or pipes and extracts heat from the combustion products. The heated air may then be routed from the furnace system **70** to the ductwork **68** for heating the residence **52**.

(35) FIG. **4** is an embodiment of a vapor compression system **72** that can be used in any of the systems described above. The vapor compression system **72** may circulate a refrigerant through a circuit starting with a compressor **74**. The circuit may also include a condenser **76**, an expansion valve(s) or device(s) **78**, and an evaporator **80**. The vapor compression system **72** may further include a control panel **82** that has an analog to digital (A/D) converter **84**, a microprocessor **86**, a non-volatile memory **88**, and/or an interface board **90**. The control panel **82** and its components may function to regulate operation of the vapor compression system **72** based on feedback from an operator, from sensors of the vapor compression system **72** that detect operating conditions, and so forth.

(36) In some embodiments, the vapor compression system **72** may use one or more of a variable speed drive (VSDs) **92**, a motor **94**, the compressor **74**, the condenser **76**, the expansion valve or device **78**, and/or the evaporator **80**. The motor **94** may drive the compressor **74** and may be powered by the variable speed drive (VSD) **92**. The VSD **92** receives alternating current (AC) power having a particular fixed line voltage and fixed line frequency from an AC power source, and provides power having a variable voltage and frequency to the motor **94**. In other embodiments, the motor **94** may be powered directly from an AC or direct current (DC) power source. The motor **94** may include any type of electric motor that can be powered by a VSD or directly from an AC or DC power source, such as a switched reluctance motor, an induction motor, an electronically commutated permanent magnet motor, or another suitable motor.

(37) The compressor **74** compresses a refrigerant vapor and delivers the vapor to the condenser **76** through a discharge passage. In some embodiments, the compressor **74** may be a centrifugal compressor. The refrigerant vapor delivered by the compressor **74** to the condenser **76** may transfer heat to a fluid passing across the condenser **76**, such as ambient or environmental air **96**. The

refrigerant vapor may condense to a refrigerant liquid in the condenser **76** as a result of thermal heat transfer with the environmental air **96**. The liquid refrigerant from the condenser **76** may flow through the expansion device **78** to the evaporator **80**.

(38) The liquid refrigerant delivered to the evaporator **80** may absorb heat from another air stream, such as a supply air stream **98** provided to the building **10** or the residence **52**. For example, the supply air stream **98** may include ambient or environmental air, return air from a building, or a combination of the two. The liquid refrigerant in the evaporator **80** may undergo a phase change from the liquid refrigerant to a refrigerant vapor. In this manner, the evaporator **80** may reduce the temperature of the supply air stream **98** via thermal heat transfer with the refrigerant. Thereafter, the vapor refrigerant exits the evaporator **80** and returns to the compressor **74** by a suction line to complete the cycle.

(39) In some embodiments, the vapor compression system **72** may further include a reheat coil in addition to the evaporator **80**. For example, the reheat coil may be positioned downstream of the evaporator relative to the supply air stream **98** and may reheat the supply air stream **98** when the supply air stream **98** is overcooled to remove humidity from the supply air stream **98** before the supply air stream **98** is directed to the building **10** or the residence **52**.

(40) Any of the features described herein may be incorporated with the HVAC unit **12**, the residential heating and cooling system **50**, or other HVAC systems. Additionally, while the features disclosed herein are described in the context of embodiments that directly heat and cool a supply air stream provided to a building or other load, embodiments of the present disclosure may be applicable to other HVAC systems as well. For example, the features described herein may be applied to mechanical cooling systems, free cooling systems, chiller systems, or other heat pump or refrigeration applications.

(41) An HVAC system, such as the HVAC unit **12**, may utilize a reheat heat exchanger to enable dehumidification of an air flow conditioned by the HVAC system. Moreover, the HVAC system may be a heat pump having a refrigerant or vapor compression circuit configured to operate in a heating mode or in a cooling mode. To this end, embodiments of the present disclosure include a reversible valve that may be alternated between a first configuration, which enables the heat pump to operate in the heating mode, and a second configuration, which enables the heat pump to operate in the cooling mode. In accordance with present techniques, the reversible valve also enables utilization of a reheat heat exchanger with the heat pump refrigerant circuit.

(42) In the first configuration, the reversible valve may enable a refrigerant to flow along a first flow path of the refrigerant circuit. For example, the reversible valve may receive refrigerant pressurized by a compressor of the refrigerant circuit and may direct a first portion of the refrigerant to a first heat exchanger and a second portion of the refrigerant to a reheat heat exchanger. In the first configuration of the reversible valve, the reheat heat exchanger may be used to enable dehumidification of the air flow in order to achieve a target humidity of the air flow. In the second configuration, the reversible valve may enable the refrigerant to flow along a second flow path of the refrigerant circuit. For instance, the reversible valve may receive the refrigerant pressurized by the compressor and may direct a first portion of the refrigerant to a second heat exchanger and a second portion of the refrigerant to the reheat heat exchanger. In the second configuration of the reversible valve, the reheat heat exchanger may be used in conjunction with the second heat exchanger to heat the air flow.

(43) FIG. 5 is a schematic of an embodiment of an HVAC system **150** having a refrigerant circuit **151** and a reversible valve **152** configured to selectively enable operation of the HVAC system **150** in a heating mode and in a cooling mode with the refrigerant circuit **151**. As such, the HVAC system **150** may be considered a heat pump system. The reversible valve **152** may include a body **154** and a slide **156** disposed within the body **154**. The slide **156** may be configured to move within the body **154**, such as in a first direction **158** and in a second direction **160**. Based on a position of the slide **156** within the body **154**, the HVAC system **150** may operate in the heating mode or in the

cooling mode. In particular, the operating mode of the HVAC system **150** may be based on the flow of refrigerant through certain ports of the reversible valve **152** and thus based on a flow of the refrigerant along a particular flow path of the refrigerant circuit **151**.

(44) The reversible valve **152** includes an inlet **162** configured to receive refrigerant discharged by a compressor, such as compressor **74**, disposed along the refrigerant circuit **151**. The reversible valve **152** also includes a first heat exchanger port **164** that is fluidly coupled to a first heat exchanger **166** of the refrigerant circuit **151** and a second heat exchanger port **168** that is fluidly coupled to a second heat exchanger **170** of the refrigerant circuit **151**. The reversible valve **152** may further include a first reheat port **172** that is fluidly coupled to a reheat heat exchanger **174** of the refrigerant circuit **151**, a second reheat port **176** that is fluidly coupled to the reheat heat exchanger **174**, and an outlet **178** configured to direct refrigerant toward the compressor **74**, such as via an accumulator **180** configured to deliver refrigerant vapor to a suction side of the compressor **74**. Although the first heat exchanger port **164**, the second heat exchanger port **168**, the first reheat port **172**, the second reheat port **176**, and the outlet **178** are aligned with one another along a common side of the body **154** in the illustrated embodiment, the ports **164**, **168**, **172**, **176**, **178** may be positioned relative to one another in any other suitable arrangement, such as in a staggered arrangement about the body **154**. Additionally or alternatively, the HVAC system **150** may include a pressure regulator **177**, which may be a crankcase pressure regulator, configured to regulate a flow of refrigerant into the compressor **74**, such as during a startup of the operation of the compressor **74**. In certain embodiments, the pressure regulator **177** may be integrated with the reversible valve **152** rather than a separate component.

(45) The HVAC system **150** may operate the refrigerant circuit **151** in either the heating mode or the cooling mode based on the flow of refrigerant through the reversible valve **152**. For example, the slide **156** may include a portion **179** having a generally U-shaped cross-section that is configured to direct the flow of refrigerant through the reversible valve **152** and through the refrigerant circuit **151** in a particular manner and/or along particular flow paths of the refrigerant circuit **151**. In some embodiments, the portion **179** may have a hemispherical, conical, cup-shaped, or other suitable geometry.

(46) In the illustrated embodiment, the reversible valve **152** is in a heating or first configuration, and the slide **156** is in a heating mode or first position to enable operation of the refrigerant circuit **151** in the heating mode. In the heating mode, refrigerant that has been pressurized and heated by the compressor **74** is directed into the reversible valve **152** via the inlet **162**. Due to the position of the slide **156**, the heated refrigerant may then be discharged to the reheat heat exchanger **174** via the first reheat port **172**. In some embodiments, the refrigerant circuit **151** may include a first three-way valve **181** configured to regulate refrigerant flow from the first reheat port **172** to the reheat heat exchanger **174** and/or to the second reheat port **176** of the reversible valve **152**. With the slide **156** in the heating mode position, refrigerant may also be discharged from the reversible valve **152** via the first heat exchanger port **164** toward the first heat exchanger **166**. For instance, a first portion of the heated refrigerant may be discharged through the first heat exchanger port **164** toward the first heat exchanger **166**, and a second portion of the heated refrigerant may be discharged through the first reheat port **172** to the reheat heat exchanger **174**. As shown, in the heating mode position, the slide **156** blocks refrigerant flow from the inlet **162** to the second heat exchanger port **168**, the second reheat port **176**, and the outlet **178**.

(47) In some embodiments, the first heat exchanger **166** and the reheat heat exchanger **174** may each be placed in a heat exchange relationship with an entering air flow **182**, which may include a return air flow from the building serviced by the HVAC system **150** and/or an outdoor air flow from an outdoor environment. The entering air flow **182** may be directed across the first heat exchanger **166**, where heat may transfer from the first portion of the heated refrigerant to the entering air flow **182** in order to increase the temperature of the entering air flow **182** and decrease the temperature of the first portion of the heated refrigerant. Since the temperature of the first

portion of the heated refrigerant decreases in the first heat exchanger **166** in the heating mode of the HVAC system **150**, the first heat exchanger **166** may operate as a condenser in the heating mode of the HVAC system **150**.

(48) The entering air flow **182** may then be directed through the reheat heat exchanger **174**, where heat may transfer from the second portion of the heated refrigerant to the entering air flow **182** to further increase the temperature of the entering air flow **182**. The entering air flow **182** may then be delivered to the building as a supply air flow **184** to heat a space within the building. It should be noted that the heating provided by both the first heat exchanger **166** and the reheat heat exchanger **174** may sufficiently increase the temperature of the entering air flow **182** such that the entering air flow **182** is suitable to deliver to the building as the supply air flow **184** and heat the space within the building by a desired amount. In other words, an additional heat source may not be incorporated into the HVAC system **150** to provide additional heat and further increase the temperature of the entering air flow **182**. Alternatively, if an additional heat source is included, the first heat exchanger **166** and the reheat heat exchanger **174** may substantially increase the temperature of the entering air flow **182**, such that the additional heat source may be operated at a lower power or operating level, thereby reducing operating costs. Thus, utilization of the reheat heat exchanger **174** in conjunction with the first heat exchanger **166** to heat the entering air flow **182** may reduce a cost associated with operating the additional heat source and/or the HVAC system **150**.

(49) The refrigerant circuit **151** of the HVAC system **150** may also include a first expansion device **186** and a second expansion device **188**. For example, the expansion devices **186**, **188** may each be expansion valves. The expansion devices **186**, **188** may each decrease a pressure of the refrigerant, thereby decreasing a temperature of the refrigerant as well. In the illustrated embodiment, the first heat exchanger **166** discharges the first portion of the refrigerant through the first expansion device **186**. The first portion of the refrigerant then flows through a second three-way valve **190** where the first portion is combined with the second portion of the refrigerant discharged from the reheat heat exchanger **174**. The second three-way valve **190** then directs the combined refrigerant to the second expansion device **188**, which further expands the refrigerant. In additional or alternative embodiments, instead of the second three-way valve **190**, a check valve and/or a two-way valve may be used to enable flow of refrigerant from the reheat heat exchanger **174** to the second expansion device **188** and to block refrigerant flow from the first expansion device **186** into the reheat heat exchanger **174**. Downstream of the second expansion device **188**, the refrigerant is directed through the second heat exchanger **170**, where the refrigerant may be placed in a heat exchange relationship with an ambient air flow **185** from an ambient environment. Heat may transfer between the ambient environment and the refrigerant and may thereby heat the refrigerant in the heating mode. Since the temperature of the refrigerant increases in the second heat exchanger **170** in the heating mode of the HVAC system **150**, the second heat exchanger **170** may operate as an evaporator in the heating mode. The second heat exchanger **170** may direct the refrigerant toward the second heat exchanger port **168** and into the reversible valve **152**. In the reversible valve **152**, the slide **156** may direct the refrigerant from the second heat exchanger port **168** to the outlet **178**. That is, in the heating mode, the slide **156** blocks refrigerant flow from the second heat exchanger port **168** to the first heat exchanger port **164** and the first reheat port **172** and enables the refrigerant to flow from the second heat exchanger port **168** to the outlet **178**. After discharging from the reversible valve **152** via the outlet **178**, the refrigerant is directed toward the accumulator **180** and the compressor **74** to be pressurized and further heated.

(50) In certain implementations, the first heat exchanger port **164**, the second heat exchanger port **168**, the first reheat port **172**, and the second reheat port **176** may each include a valve **192** configured to enable control of an amount, such as a volumetric flowrate, of refrigerant flow through the respective ports **164**, **168**, **172**, **176**. Each valve **192** may be controlled via a controller **194** of the HVAC system **150**. For example, the controller **194** may include a memory **196** and a processor **198**. The memory **196** may be a mass storage device, a flash memory device, removable

memory, or any other non-transitory computer-readable medium that includes instructions for the processor **198** to execute. The memory **196** may also include volatile memory such as randomly accessible memory (RAM) and/or non-volatile memory such as hard disc memory, flash memory, and/or other suitable memory formats. The processor **198** may execute the instructions stored in the memory **196**, such as to control an amount of refrigerant flowing through each valve **192**. As an example, the controller **194** may receive an input indicative of an operating parameter of the HVAC system **150**, such as a desired temperature of the space within the building, a current temperature of the space within the building, a current humidity of the space within the building, a temperature of the entering air flow **182**, a humidity of the entering air flow **182**, a target temperature of the supply air flow **184**, a target humidity of the supply air flow **184**, another suitable parameter, or any combination thereof.

(51) In the heating mode, the controller **194** may set the position of a first valve **192A** of the first heat exchanger port **164** and the position of a second valve **192B** of the first reheat port **172** to control the amount of refrigerant in the first portion discharged toward the first heat exchanger **166** and the amount of refrigerant in the second portion discharged toward the reheat heat exchanger **174**, respectively. Moreover, the controller **194** may set the position of the first three-way valve **181** to enable the heated refrigerant to flow from the second valve **192B** to the reheat heat exchanger **174** and to block the heated refrigerant from flowing from the second valve **192B** toward the second reheat port **176**. The controller **194** may also set the position of the second three-way valve **190** to enable the refrigerant to flow from the first heat exchanger **166** and from the reheat heat exchanger **174** to the second expansion device **188**.

(52) In one example, the controller **194** may set the position of the first valve **192A** and the position of the second valve **192B** to enable substantially the same amount of heated refrigerant to flow from the reversible valve **152** to the first heat exchanger **166** and to the reheat heat exchanger **174**. As such, the first heat exchanger **166** and the reheat heat exchanger **174** may have similar heating capacities. In another example, the controller **194** may set the position of the first valve **192A** and the position of the second valve **192B** such that an increased amount of heated refrigerant flows through the reheat heat exchanger **174** and a decreased amount of heated refrigerant flows through the first heat exchanger **166**. As such, the reheat heat exchanger **174** may have a greater heating capacity than that of the first heat exchanger **166**. In a further example, the controller **194** may close the second valve **192B**, such that substantially all of the heated refrigerant flows from the reversible valve **152** through the first heat exchanger **166**, and substantially none of the heated refrigerant flows through the reheat heat exchanger **174**. Thus, the reheat heat exchanger **174** may not be used to heat the entering air flow **182**, and operation of the reheat heat exchanger **174** may be suspended or disabled. In some embodiments, remaining refrigerant contained within the reheat heat exchanger **174** may be discharged when operation of the reheat heat exchanger **174** is suspended or disabled. For instance, the controller **194** may position each of the first three-way valve **181** and the fourth valve **192D** to enable any remaining refrigerant in the reheat heat exchanger **174** to flow into the reversible valve **152** via the second reheat port **176**. For example, in the illustrated heating mode position of the reversible valve **152**, a suction pressure generated by the compressor **74** may draw any remaining refrigerant in the reheat heat exchanger **174** through the first three-way valve **181**, through the fourth valve **192D**, and through the outlet **178**. As such, no refrigerant may remain in the reheat heat exchanger **174**.

(53) Furthermore, in the heating mode, the controller **194** may set the position of a third valve **192C** of the second heat exchanger port **168** to enable refrigerant to flow from the second heat exchanger **170** into the reversible valve **152**. The controller **194** may also set the position of a fourth valve **192D** of the second reheat port **176** to block the refrigerant from flowing toward the reheat heat exchanger **174** from the reversible valve **152**. The suction pressure generated by the compressor **74** may also cause substantially all of the refrigerant flowing from the second heat exchanger **170** into the reversible valve **152** to discharge through the outlet **178**.

(54) The HVAC system **150** may also have a hot gas bypass mode to enable hot gas bypass flow of the refrigerant. In the hot gas bypass mode, at least a portion of the refrigerant is directed through the refrigerant circuit **151** while bypassing the reheat heat exchanger **174** and/or the first heat exchanger **166**. Instead, the reversible valve **152** discharges at least a portion of the refrigerant to flow along a flow path that does not include the reheat heat exchanger **174** or the first heat exchanger **166**. Generally, the hot gas bypass flow enables the compressor **74** to operate at low operating modes. In low load heating mode, there is a low demand for heating, and the HVAC system **150** may operate to slightly increase the temperature of the entering air flow **182**. For example, the second portion of the heated refrigerant may be directed to flow along a flow path that does not include the reheat heat exchanger **174**. The second portion of the heated refrigerant flows from the first reheat port **172** to the first three-way valve **181**, which directs the second portion of the heated refrigerant to the second reheat port **176** via a first bypass flow path **200**. As an example, the controller **194** may position the three-way valve **181** to enable refrigerant flow from the first reheat port **172** to the second reheat port **176** and to block refrigerant flow from the first reheat port **172** to the reheat heat exchanger **174**. Furthermore, the controller **194** may position the fourth valve **192D** to enable the heated refrigerant to flow through the second reheat port **176** and to mix with refrigerant entering the reversible valve **152** through the second heat exchanger port **168** and flowing toward the outlet **178**. As no refrigerant is directed to the reheat heat exchanger **174**, the reheat heat exchanger **174** does not provide any additional heating capabilities. In this manner, the HVAC system **150** may operate to deliver the supply air **184** without overheating the entering air flow **182** and/or without suspending the operation of the compressor **74**. Indeed, in the hot gas bypass mode, the compressor **74** may operate at a reduce capacity while the HVAC system **150** satisfies a low load demand, thereby reducing operating costs of the HVAC system **150**.

(55) FIG. **6** is a schematic of an embodiment of the HVAC system **150**, in which the reversible valve **152** is in a cooling or second configuration, and the slide **156** is in a cooling mode or second position to enable operation of the refrigerant circuit **151** in the cooling and/or dehumidification mode. During the cooling and/or dehumidification mode, the reversible valve **152** receives refrigerant that has been pressurized and heated by the compressor **74** via the inlet **162**. The position of the slide **156** enables a first portion of the heated refrigerant to be discharged through the second heat exchanger port **168** to the second heat exchanger **170**. Furthermore, a second portion of the heated refrigerant is discharged from the reversible valve **152** through the second reheat port **176** toward the first three-way valve **181** and then to the reheat heat exchanger **174**. To this end, in the cooling mode, the slide **156** blocks refrigerant flow from the inlet **162** to the first heat exchanger port **164**, the first reheat port **172**, and the outlet **178**.

(56) The first portion of the heated refrigerant directed to the second heat exchanger **170** may be placed in a heat exchange relationship with the ambient air flow **185**, which may cause heat to transfer from the heated refrigerant to the ambient air flow **185**. Thus, the first portion of the heated refrigerant may be cooled, and the second heat exchanger **170** may operate as a condenser in the cooling mode of the HVAC system **150**. The second heat exchanger **170** may discharge the first portion of the heated refrigerant to the second expansion device **188**, which may expand and further cool the first portion of the refrigerant. The first portion of the heated refrigerant may then be directed to the second three-way valve **190**.

(57) The second portion of heated refrigerant that is directed through the reheat heat exchanger **174** is utilized therein for reheat and dehumidification of the entering air **182**, as discussed below. After discharging from the reheat heat exchanger **174**, the second portion of the refrigerant is directed to the second three-way valve **190**. However, as the second portion of the heated refrigerant has not been expanded by an expansion device or valve before entering the second three-way valve **190**, the second portion of the heated refrigerant may have a temperature higher than the first portion of the heated refrigerant received by the second three-way valve **190** from the second expansion device **188**. Nevertheless, the first and second portions of the heated refrigerant may combine with

one another at the second three-way valve **190** and may be directed to the first expansion device **186**. At the first expansion device **186**, the combined refrigerant may expand and cool and may then be directed to the first heat exchanger **166**. In this manner, the refrigerant flowing through the first heat exchanger **166** may be at a substantially lower temperature than the temperature of the second portion of the heated refrigerant flowing through the reheat heat exchanger **174**. The first heat exchanger **166** may then direct the refrigerant through the first heat exchanger port **164** and into the reversible valve **152**, where the portion **179** of the slide **156** may direct the refrigerant out of the slide **156** via the outlet **178** and toward the compressor **74** to be pressurized. Thus, in the cooling mode, the slide **156** blocks refrigerant flow from the first heat exchanger port **164** to the second heat exchanger port **168** and to the second reheat port **176**.

(58) The first heat exchanger **166** and the reheat heat exchanger **174** may each place the refrigerant in a heat exchange relationship with the entering air flow **182** in the cooling mode. As noted above, the temperature of the combined refrigerant in the first heat exchanger **166** may be substantially lower than the temperature of the first portion of the heated refrigerant in the reheat heat exchanger **174**. Thus, heat may transfer from the entering air flow **182** to the refrigerant in the first heat exchanger **166**, thereby cooling the entering air flow **182** and heating the refrigerant. As such, the first heat exchanger **166** may operate as an evaporator in the cooling mode of the HVAC system **150**. Cooling the entering air flow **182** may condense and remove moisture from the entering air flow **182**, thereby reducing a humidity of the entering air flow **182**, such as toward a target humidity. The cooled entering air flow **182** may then be directed to the reheat heat exchanger **174**, where heat may transfer from refrigerant to the entering air flow **182** to heat the entering air flow **182**, such as toward a target temperature. The entering air flow **182** may then be delivered to the building as the supply air flow **184**.

(59) In the cooling mode, the entering air flow **182** may be conditioned via the first heat exchanger **166** and/or the reheat heat exchanger **174** to achieve a target humidity and/or temperature of the supply air flow **184** in order to provide a more comfortable supply air flow **184** to the space within the building. For example, the first heat exchanger **166** may be used to reduce the temperature of the entering air flow **182** to a first target temperature in order to remove a certain amount of moisture from the entering air flow **182** and to achieve a target humidity of the entering air flow **182**. However, the first target temperature may be unsuitable, such as too low, for the supply air flow **184**. In other words, by cooling and dehumidifying the entering air flow **182** with the first heat exchanger **166**, the temperature of the entering air flow **182** may fall below a desired temperature to cool the space within the building. For this reason, the reheat heat exchanger **174** may be utilized to increase the temperature of the entering air flow **182** to a second target temperature, which may be a more suitable temperature for the supply air flow **184**, while maintaining the humidity of the entering air flow **182** at the target humidity. Thus, using the reheat heat exchanger **174** may enable the HVAC system **150** to deliver a more comfortable supply air flow **184** to the building.

(60) In some embodiments, the controller **194** may set the position of the third valve **192C** of the second heat exchanger port **168** and the position of the fourth valve **192D** of the second reheat port **176** to control the amount of heated refrigerant in the first portion discharged toward the second heat exchanger **170** and the amount of heated refrigerant in the second portion discharged toward the reheat heat exchanger **174**, respectively. The controller **194** may also set the position of the first three-way valve **181** to enable the heated refrigerant to flow from the fourth valve **192D** to the reheat heat exchanger **174** and to block the heated refrigerant from flowing from the fourth valve **192D** toward the first reheat port **172**. Further, the controller **194** may set the position of the second three-way valve **190** to enable the refrigerant to flow from the second heat exchanger **170** and from the reheat heat exchanger **174** to the first expansion device **186**.

(61) The controller **194** may set the position of the third valve **192C** and the position of the fourth valve **192D** to enable the reheat heat exchanger **174** to provide a target amount of reheating and/or to enable the first heat exchanger **166** to provide a target amount of cooling and/or

dehumidification. For example, if increased reheating by the reheat heat exchanger **174** is desired, the controller **194** may set the positions of the third valve **192C** and the fourth valve **192D** such that the amount of heated refrigerant discharged to the reheat heat exchanger **174** is increased and the amount of heated refrigerant discharged to the second heat exchanger **170** is decreased. If no reheating by the reheat heat exchanger **174** is desirable, the controller **194** may close the fourth valve **192D**, such that substantially all of the heated refrigerant is discharged from the reversible valve **152** to the second heat exchanger **170**. Thus, no refrigerant may flow through the reheat heat exchanger **174**, and the controller **194** may suspend or disable operation of the reheat heat exchanger **174**. In some embodiments, when the reheat heat exchanger **174** is not in operation, the reheat heat exchanger **174** may discharge remaining refrigerant within the reheat heat exchanger **174** from previous operation. For example, the controller **194** may set the first three-way valve **181** and the second valve **192B** to enable the remaining refrigerant in the reheat heat exchanger **174** to flow into the reversible valve **152** via the first reheat port **172**, such as due to a suction pressure generated by the compressor **74**, as similarly described above.

(62) In the cooling and/or dehumidification mode, the controller **194** may set the position of the first valve **192A** of the first heat exchanger port **164** to enable refrigerant to flow from the first heat exchanger **166** into the reversible valve **152**. Also, the controller **194** may set the position of the second valve **192B** of the first reheat port **172** to block refrigerant flow toward the reheat heat exchanger **174** from the first reheat port **172**. Thus, substantially all of the refrigerant flowing from the first heat exchanger **166** into the reversible valve **152** is discharged through the outlet **178**.

(63) The HVAC system **150** may also operate in a hot gas bypass mode in the during cooling and/or dehumidification operations. That is, the HVAC system **150** may enable hot gas bypass flow of the refrigerant in the cooling mode to enable the compressor **74** to operate in a low load cooling mode. In the low load cooling mode, there is a low demand for cooling and/or a low demand for dehumidification, and the HVAC system **150** may operate to slightly reduce the temperature of the entering air flow **182** without substantially changing the humidity of the entering air flow **182**. For example, the HVAC system **150** may be configured to discharge the second portion of the heated refrigerant from the reversible valve **152** to flow along a flow path of the refrigerant circuit **151** that bypasses the reheat heat exchanger **174**.

(64) In one embodiment, the controller **194** may set the position of the first three-way valve **181** to enable the second portion of the heated refrigerant to flow from the second reheat port **176** to the first three-way valve **181** and from the first three-way valve **181** to the first reheat port **172** while blocking the heated refrigerant from flowing from the second reheat port **176** to reheat heat exchanger **176**. The controller **194** may further set the position of the second valve **192B** to enable the heated refrigerant to flow from the first three-way valve **181** through the second heat exchanger port **172** along a second bypass flow path **220**. The heated refrigerant may then combine with refrigerant flowing through the first heat exchanger port **164** from the first heat exchanger **166**, and the combined refrigerant may flow toward the outlet **178**. By increasing the amount of heated refrigerant directed through the second reheat port **176** to bypass the reheat heat exchanger **174**, less heated refrigerant may be directed through the second heat exchanger **170** and then to the first heat exchanger **166**. Thus, the cooling capacity of the first heat exchanger **166** may decrease. In other words, with a reduced refrigerant flow through the second heat exchanger **170** and the first heat exchanger **166**, and with no refrigerant flow through the reheat heat exchanger **174**, the HVAC system **150** may be operated to slightly cool the entering air flow **182** without substantially changing the humidity of the entering air flow **182** and/or without suspending operation of the compressor **74**. Indeed, in the hot gas bypass mode, the compressor **74** may operate at a reduced capacity while the HVAC system **150** satisfies a low load demand, thereby reducing operating costs of the HVAC system **150**.

(65) FIG. 7 is a schematic of an embodiment of the HVAC system **150** having the reversible valve **152** and an additional heat exchanger **250** shared by an additional HVAC system **252**, which may

be another vapor compression system or a second refrigerant circuit **251**. The additional HVAC system **252** may be considered a part of or integrated with the HVAC system **150**. In some embodiments, the additional heat exchanger **250** may be integrated or packaged with the first heat exchanger **166**. For example, the first heat exchanger **166** and the additional heat exchanger **250** may each be a separate section of a common heat exchanger, which may have a face split arrangement, a row split arrangement, an interlaced split arrangement, another suitable arrangement, or any combination thereof that fluidly separates the first heat exchanger **166** from the additional heat exchanger **250**. In other words, the first heat exchanger **166** may flow a first refrigerant circulated within the refrigerant circuit **151**, and the additional HVAC system **252** may flow a second refrigerant circulated within the second refrigerant circuit **251**.

(66) The HVAC system **252** may condition the entering air flow **182** via the second refrigerant. By way of example, the additional HVAC system **252** may direct the second refrigerant through the additional heat exchanger **250**, which places the second refrigerant in a heat exchange relationship with the entering air flow **182**. In the illustrated embodiment, the entering air flow **182** is directed through the first heat exchanger **166** to transfer heat with the refrigerant from the HVAC system **150**, then through the additional heat exchanger **250** to transfer heat with the second refrigerant from the additional HVAC system **252**, and then through the reheat heat exchanger **174** to transfer additional heat with the refrigerant from the HVAC system **150**.

(67) The additional HVAC system **252** may also be configured to operate in a cooling mode and in a heating mode. For example, the additional HVAC system **252** may be a heat pump. In some embodiments, both the HVAC system **150** and the additional HVAC system **252** may be operated in the cooling mode, such that the additional heat exchanger **250** provides supplemental cooling of the entering air flow **182**. In alternative embodiments, the HVAC system **150** may be operated in the cooling mode, and the additional HVAC system **252** may be operated in the heating mode. In such embodiments, the additional heat exchanger **250** may provide supplemental reheating of the entering air flow **182** after the entering air flow **182** has been cooled via the first heat exchanger **166**. In further embodiments, both the HVAC system **150** and the additional HVAC system **252** may be operated in the heating mode. Thus, the first heat exchanger **166**, the reheat heat exchanger **174**, and the additional heat exchanger **250** may each heat the entering air flow **182**.

(68) FIG. **8** is a schematic of an embodiment of the HVAC system **150** having the reversible valve **152** positioned to operate the HVAC system **150** in the cooling mode. In the illustrated embodiment, the reversible valve **152** includes a single reheat port **300**, instead of the first reheat port **172** and the second reheat port **176**. In the illustrated embodiment, in which the reversible valve **152** is in the cooling configuration and the slide **156** is in the cooling mode position, heated refrigerant flowing into the reversible valve **152** from the compressor **74** may be discharged to the second heat exchanger **170** via the second heat exchanger port **168** and/or to the reheat heat exchanger **174** via the reheat port **300**. Conversely, in the heating mode, in which the reversible valve **152** is in the heating configuration and the slide **156** is in the heating position, heated refrigerant flowing into the reversible valve **152** may be discharged to the first heat exchanger **166** via the first heat exchanger port **164** and/or to the reheat heat exchanger **174** via the reheat port **300**. In other words, in both the heating mode and the cooling mode, heated refrigerant may be discharged out of the reversible valve **152** through the reheat port **300** toward the reheat heat exchanger **174**. Although the reheat port **300** is aligned with and adjacent to the inlet **162** in the illustrated embodiment, the reheat port **300** may be positioned in any suitable manner along the body **154** of the reversible valve **152**.

(69) It should be noted that, in the illustrated embodiment, operation of the HVAC system **150** in the heating mode and in the cooling mode may be similar to the techniques described above with reference to FIGS. **5** and **6**. For instance, a fifth valve **192E** may be disposed in or along the reheat port **300** and may be controlled by the controller **194** to adjust an amount, such as a volumetric flowrate, of heated refrigerant directed to the reheat heat exchanger **174**. By way of example, in the

cooling mode, the controller **194** may set the position of the third valve **192C** of the second heat exchanger port **168** and the position of the fifth valve **192E** of the reheat port **300** to control the amount of heated refrigerant flowing to the second heat exchanger **170** relative to the amount of heated refrigerant flowing to the reheat heat exchanger **174**. In the heating mode, the controller **194** may set the position of the first valve **192A** of the first heat exchanger port **164** and the position of the fifth valve **192E** of the reheat port **300** to control the amount of heated refrigerant flowing to the first heat exchanger **166** relative to the amount of heated refrigerant flowing to the reheat heat exchanger **174**. It should also be noted that, since the reheat port **300**, rather than separate reheat ports **172**, **176**, is used to direct refrigerant to the reheat heat exchanger **174**, the HVAC system **150** may not have the first three-way valve **181**. In certain embodiments, the reheat port **300** may be fluidly coupled to the compressor **74** to enable refrigerant to be directed from the reheat port **300** toward the compressor **74** and enable features of hot gas bypass similarly discussed above.

(70) FIG. **9** is a schematic of an embodiment of the reversible valve **152**, which is fluidly coupled to a pilot valve **350** configured to adjust the position of the slide **156** within the body **154** of the reversible valve **152**. The pilot valve **350** includes a first port **352** that is fluidly coupled to the inlet **162** of the reversible valve **152** via a first pilot line **354**. The pilot valve **350** also includes a second port **356** that is fluidly coupled to the outlet **178** of the reversible valve **152** via a second pilot line **358**. The pilot valve **350** may further include a third port **360**, which is fluidly coupled to a first chamber **362** in the body **154** of the reversible valve **152** via a third pilot line **364**, and a fourth port **366**, which is fluidly coupled to a second chamber **368** opposite the first chamber **362** in the body **154** of the reversible valve **152** via a fourth pilot line **370**.

(71) The first chamber **362** may enable fluid flow from the pilot valve **350** to a first side **372** of the slide **156**, and the second chamber **368** may enable fluid flow from the pilot valve **350** to a second side **374** of the slide **156**. In the illustrated embodiment, the reversible valve **152** is in the heating configuration to enable the HVAC system **150** to operate in the heating mode. To set the reversible valve **152** in the heating configuration, the controller **194** may actuate the pilot valve **350** to drive a pilot slide **376** in the second direction **160** to a first pilot valve position. In some embodiments, the pilot valve **350** may be a solenoid valve, and the controller **194** may transmit an electrical current through a coil of the solenoid valve to create a magnetic field that drives the pilot slide **376** in the second direction **160**. In additional or alternative embodiments, the pilot valve **350** may include an actuator that the controller **194** may activate to drive the pilot slide **376** in the second direction **160**.

(72) While the pilot slide **376** is in the first pilot valve position, some refrigerant pressurized by the compressor **74** may flow from the inlet **162** through the first pilot line **354** in a first flow direction **378** and into the pilot valve **350** via the first port **352**. The pilot slide **376** may then direct the pressurized refrigerant in a second flow direction **380** out of the pilot valve **350** via the fourth port **366**, through the fourth pilot line **370**, and in a third flow direction **382** into the second chamber **368**. That is, the pilot slide **376** blocks the pressurized refrigerant from flowing from the first port **352** through the second port **356** or the third port **360** when in the first pilot valve position.

Additionally, some of the refrigerant, which may have been depressurized via the expansion devices **186**, **188**, flowing out of the body **154** of the reversible valve **152** via the outlet **178** may flow through the second pilot line **358** in a fourth flow direction **384** and into the pilot valve **350** via the second port **356**. The position of the pilot slide **376** may block the refrigerant from flowing from the second port **356** to the first port **352** or the fourth port **366**. Instead, the pilot slide **376** may direct the refrigerant in a fifth flow direction **386** to flow out of the pilot valve **350** via the third port **360**, through the third pilot line **364**, and in a sixth flow direction **388** into the first chamber **362**. The pressure of the refrigerant directed into the second chamber **368** from the compressor **74** via the pilot valve **350** may be greater than the pressure of the refrigerant directed into the first chamber **362** from the outlet **178** of the reversible valve **152**. As a result, the refrigerant in the second chamber **368** may exert a greater force onto a first slide piston **390** disposed on the second side **374** of the slide **156** than a force exerted on a second slide piston **392**

disposed on the first side **372** of the slide **156** by the refrigerant in the first chamber **362**. Therefore, the slide **156** is driven in the second direction **160** to the heating position, which enables the reversible valve **152**, and thus the HVAC system **150**, to operate in the heating mode.

(73) FIG. **10** is a schematic of an embodiment of the reversible valve **152** fluidly coupled to the pilot valve **350** and in the cooling configuration. To this end, the controller **194** may actuate the pilot slide **376** to move in the first direction **158** into a second pilot valve position. While the pilot slide **376** is in the second pilot valve position, some refrigerant pressurized by the compressor **74** may flow from the inlet **162** of the reversible valve **152**, through the first pilot line **354** in the first flow direction **378**, and into the pilot valve **350** via the first port **352**. The pilot slide **376** may then direct the pressurized refrigerant in a seventh flow direction **420** out of the pilot valve **350** via the third port **360**, through the third pilot line **364**, and then to the first chamber **362**. Thus, in the second pilot valve position, the pilot slide **376** blocks the pressurized refrigerant from flowing from the first port **352** to the second port **356** or the fourth port **366**. Some of the refrigerant flowing out of the outlet **178** may flow through the second pilot line **358** in the fourth flow direction **384** into the pilot valve **350** via the second port **356**, and the pilot slide **376** may block refrigerant from flowing from the second port **356** to the first port **352** or the third port **360**. Thus, the refrigerant may be directed in an eighth flow direction **422** to flow out of the pilot valve **350** via the fourth port **366**. The refrigerant may then flow through the fourth pilot line **370** and into the second chamber **368**. In the second pilot valve position, the pressure of the refrigerant directed into the first chamber **362** may be greater than the pressure of the refrigerant directed into the second chamber **368**. Thus, the refrigerant in the first chamber **362** may impart a greater force onto the second slide piston **392** on the first side **372** of the slide **156** than a force imparted by the refrigerant onto the first slide piston **390** in the second chamber **368**. As such, the slide **156** is driven in the first direction **158** to the cooling mode position, thereby enabling the reversible valve **152** and the HVAC system **150** to operate in the cooling mode.

(74) The present disclosure is directed to an HVAC system, such as a heat pump, having a reversible valve and a reheat heat exchanger to facilitate conditioning of an air flow delivered to a building. A first configuration of the reversible valve may enable the heat pump to operate in a heating mode, and a second configuration of the reversible valve may enable the heat pump to operate in a cooling mode. In both the heating mode and the cooling mode, the reversible valve may receive pressurized refrigerant from a compressor of the heat pump and may discharge the pressurized refrigerant to the reheat heat exchanger of the heat pump. In the heating mode, a heat exchanger of the heat pump may be operated to heat the air flow, and the reheat heat exchanger may provide further heating of the air flow and increase the efficiency of the heat pump to heat the building. In the cooling mode, the heat exchanger of the heat pump may be operated to cool the air flow, and the reheat heat exchanger may provide reheating capabilities that enable greater temperature and humidity control of the air flow. Thus, the reheat heat exchanger may enable the heat pump to provide a more comfortable air flow to the building. It should be noted that existing HVAC systems or heat pumps may be retrofitted with the reversible valve described in this disclosure. In other words, the reversible valve may be implemented onto an existing heat pump to enable the existing heat pump to operate in either the heating mode or the cooling mode and to direct refrigerant from the compressor to the reheat heat exchanger. The technical effects and technical problems in the specification are examples and are not limiting. It should be noted that the embodiments described in the specification may have other technical effects and can solve other technical problems.

(75) While only certain features and embodiments of the disclosure have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, including temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth without materially departing from the novel teachings and advantages of

the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure. Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual implementation may not have been described, such as those unrelated to the presently contemplated best mode of carrying out the disclosure, or those unrelated to enabling the claimed disclosure. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure, without undue experimentation.

Claims

1. A heating, ventilation, and/or air conditioning (HVAC) system, comprising: a reheat heat exchanger disposed along a refrigerant circuit, wherein the reheat heat exchanger is configured to transfer heat from a refrigerant to an air flow in a heating mode of the HVAC system and in a cooling mode of the HVAC system; a reversible valve disposed along the refrigerant circuit, wherein the reversible valve is adjustable between a first configuration corresponding to the cooling mode and a second configuration corresponding to the heating mode, the reversible valve is configured to direct the refrigerant through the refrigerant circuit in a first flow direction in the first configuration and to direct the refrigerant through the refrigerant circuit in a second flow direction in the second configuration, wherein the reversible valve comprises a reheat port, and the reversible valve is configured to direct the refrigerant toward the reheat heat exchanger via the reheat port in the first configuration and in the second configuration; and a valve associated with the reheat port, wherein the valve is configured to adjust an amount of the refrigerant directed from the reversible valve to the reheat heat exchanger via the reheat port.
2. The HVAC system of claim 1, wherein the reversible valve comprises an inlet, and the reversible valve is configured to receive the refrigerant from a compressor disposed along the refrigerant circuit via the inlet in the first configuration and in the second configuration.
3. The HVAC system of claim 2, wherein the reversible valve comprises an outlet, and the reversible valve is configured to direct the refrigerant to the compressor via the outlet in the first configuration and in the second configuration.
4. The HVAC system of claim 3, wherein the reversible valve comprises a first heat exchanger port and a second heat exchanger port, and, in the first configuration: the reversible valve is configured to receive the refrigerant from a first additional heat exchanger disposed along the refrigerant circuit via the first heat exchanger port, and the reversible valve is configured to direct the refrigerant to a second additional heat exchanger disposed along the refrigerant circuit via the second heat exchanger port.
5. The HVAC system of claim 4, wherein, in the second configuration: the reversible valve is configured to receive the refrigerant from the second additional heat exchanger via the second heat exchanger port, and the reversible valve is configured to direct the refrigerant to the first additional heat exchanger via the first heat exchanger port.
6. The HVAC system of claim 4, wherein the reheat port is disposed opposite the outlet, the first heat exchanger port, and the second heat exchanger port, relative to a body of the reversible valve.
7. The HVAC system of claim 2, wherein the inlet and the reheat port are aligned along a length of a body of the reversible valve.
8. The HVAC system of claim 1, comprising a controller communicatively coupled to the valve associated with the reheat port, wherein the controller is configured to adjust the valve to control an amount of reheat provided by the reheat heat exchanger to the air flow in the cooling mode, and the

controller is configured to adjust the valve to control an amount of supplemental heating provided by the reheat heat exchanger to the air flow in the heating mode.

9. A heating, ventilation, and/or air conditioning (HVAC) system, comprising: a refrigerant circuit; a reheat heat exchanger disposed along the refrigerant circuit, wherein the reheat heat exchanger is configured to transfer heat from a refrigerant to an air flow directed through the HVAC system in a heating mode of the HVAC system and in a cooling mode of the HVAC system; and a reversing valve disposed along the refrigerant circuit, wherein the reversing valve is configured to adjust a flow direction of the refrigerant along the refrigerant circuit, and the reversing valve comprises: an inlet configured to receive the refrigerant from a compressor in the heating mode and in the cooling mode; an outlet configured to direct the refrigerant from the reversing valve toward the compressor in the heating mode and in the cooling mode; a reheat port configured to direct the refrigerant from the reversing valve toward the reheat heat exchanger in the heating mode and in the cooling mode; a first heat exchanger port configured to receive the refrigerant from a first additional heat exchanger disposed along the refrigerant circuit in the cooling mode and configured to direct the refrigerant from the reversing valve toward the first additional heat exchanger in the heating mode; and a second heat exchanger port configured to direct the refrigerant from the reversing valve toward a second additional heat exchanger disposed along the refrigerant circuit in the heating mode and configured to receive the refrigerant from the second additional heat exchanger in the cooling mode.

10. The HVAC system of claim 9, wherein the reversing valve comprises a body and a slide disposed within the body, the slide is adjustable between a first position and a second position, the slide is configured to direct the refrigerant from the first heat exchanger port to the outlet in the first position to enable operation of the HVAC system in the cooling mode, and the slide is configured to direct the refrigerant from the second heat exchanger port to the outlet in the second position to enable operation of the HVAC system in the heating mode.

11. The HVAC system of claim 10, wherein the reversing valve is configured to direct the refrigerant from the inlet, through the body, and to the reheat port in the first position of the slide and in the second position of the slide.

12. The HVAC system of claim 9, wherein the outlet is disposed between the first heat exchanger port and the second heat exchanger port relative to a length of a body of the reversing valve.

13. The HVAC system of claim 9, wherein the inlet and the reheat port are disposed opposite the outlet relative to a body of the reversing valve.

14. A heat pump configured to operate in a heating mode and in a cooling mode, wherein the heat pump comprises: a reheat heat exchanger configured to transfer heat from a refrigerant to an air flow supplied to a conditioned space in the heating mode and in the cooling mode; a first heat exchanger disposed upstream of the reheat heat exchanger, relative to flow of the air flow through the heat pump, wherein the first heat exchanger is configured to transfer heat between the refrigerant and the air flow; a second heat exchanger configured to transfer heat between the refrigerant and an ambient air flow; and a reversible valve comprising a first heat exchanger port fluidly coupled to the first heat exchanger, a second heat exchanger port fluidly coupled to the second heat exchanger; and a reheat port fluidly coupled to the reheat heat exchanger, wherein the reheat port is configured to discharge refrigerant from the reversible valve toward the reheat heat exchanger in the heating mode and in the cooling mode.

15. The heat pump of claim 14, wherein the reheat port is disposed opposite the first heat exchanger port and the second heat exchanger port relative to a body of the reversible valve.

16. The heat pump of claim 14, wherein the reversible valve comprises: an inlet configured to receive the refrigerant from a compressor of the heat pump in the heating mode and in the cooling mode; and an outlet configured to direct the refrigerant from the reversing valve toward the compressor in the heating mode and in the cooling mode, wherein the inlet and the reheat port are disposed opposite the first heat exchanger port, the second heat exchanger port, and the outlet

relative to a body of the reversible valve.

17. The heat pump of claim 16, wherein the outlet is disposed between the first heat exchanger port and the second heat exchanger port relative to a length of the body, the reversible valve comprises a slide disposed within the body, the slide is adjustable between a first position corresponding to the cooling mode and a second position corresponding to the heating mode, and wherein: in the first position, the slide is configured to direct the refrigerant from the first heat exchanger port to the outlet, and in the second position, the slide is configured to direct the refrigerant from the second heat exchanger port to the outlet.

18. The heat pump of claim 14, comprising: a valve associated with the reheat port, wherein the valve is configured to adjust an amount of the refrigerant directed from the reversible valve to the reheat heat exchanger via the reheat port; and a controller communicatively coupled to the valve, wherein the controller is configured to: adjust the valve to control an amount of reheat provided by the reheat heat exchanger to the air flow in the cooling mode; and adjust the valve to control an amount of supplemental heating provided by the reheat heat exchanger to the air flow in the heating mode.

19. The HVAC system of claim 9, comprising a valve associated with the reheat port, wherein the valve is configured to adjust an amount of the refrigerant directed from the reversing valve toward the reheat heat exchanger via the reheat port.

20. The HVAC system of claim 19, comprising a controller communicatively coupled to the valve associated with the reheat port, wherein the controller is configured to adjust the valve to control an amount of reheat provided by the reheat heat exchanger to the air flow in the cooling mode, and the controller is configured adjust the valve to control an amount of supplemental heating provided by the reheat heat exchanger to the air flow in the heating mode.
